
SE(3)-Equivariant Flow Matching with Gaussian Process Priors for Geometric Trajectory Prediction

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Abstract

The trajectory prediction of N-body systems is of great significance and remains challenging with broad applications across various fields such as physics, chemistry and biology. Recent advances in generative models including flow matching and diffusion models have emerged as effective solutions to this problem, owing to their capacity to model the stochasticity and underlying distributions of complex system trajectories. However, existing approaches typically adopt trivial prior distributions that neglect the temporal correlations and spatial symmetries of N-body trajectories, which not only complicates the generation process but also limits model performance. To address these limitations, we propose GP-EquiFlow, an SE(3)-equivariant flow matching model incorporating vector-valued Gaussian processes. Based on observed trajectories, we employ vector-valued Gaussian processes to construct SE(3)-equivariant prior distributions, which exhibit enhanced consistency with the target data distribution in both spatial and temporal dynamics. Extensive experiments on N-body simulations and molecular dynamics demonstrate that the proposed GP-EquiFlow delivers more accurate predictions while requiring fewer sampling steps, underscoring the effectiveness of integrating Gaussian process-based SE(3)-equivariant prior distributions in geometric trajectory prediction.

1. Introduction

N-body systems, modeled by intricate dynamic interactions among particles or agents, stand as a cornerstone across diverse disciplines. Whether in simulating physical dynamics

(Stabenau & Jain, 2006; Dehnen & Read, 2011), modeling molecular dynamics (Chmiela et al., 2017; 2023), or predicting future behaviors of multi-agent systems (Amirkhani & Barshooi, 2022), the ability to reliably forecast future trajectories of such systems from historical observations directly supports scientific breakthroughs, technological innovation, and rational practical decision-making. However, trajectory prediction for N-body systems remains highly challenging, as the system dynamics are shaped by nonlinear inter-particle couplings that induce complex collective behaviors, inherently stochastic perturbations, and fundamental symmetries including rotational and translational invariance/equivariance and permutation invariance (Han et al., 2025). These properties demand predictive models that simultaneously capture temporal dependencies and strictly adhere to the intrinsic symmetries of the system.

Traditional numerical methods (Aarseth, 2003; Spurzem, 1999) are primarily based on numerical integration techniques or analytical approximations to solve deterministic and stochastic differential equations. While these methods abide by physical laws, they incur prohibitive computational costs in long-term simulations, particularly when applied to large-scale systems. Recently, generative models (Yang et al., 2023; Ho et al., 2020; Nichol & Dhariwal, 2021) have emerged as powerful tools for modeling complex trajectory distributions, leveraging their capacity to naturally accommodate the stochasticity and high dimensionality of trajectory data. Diffusion models can effectively model the distribution of system dynamics yet suffer from computationally expensive iterative sampling during inference (Shysheya et al., 2024; Bastek et al., 2025; Shehata et al., 2025; Cachay et al., 2025). By comparison, conditional flow matching (CFM) (Lipman et al., 2023; Rohbeck et al., 2025; Tong et al., 2024; Albergo et al., 2024) simplifies the generative process by constructing conditional probability paths between prior and target distributions via optimal transport, thereby enabling more efficient training and inference procedures.

Even with these advances, existing flow matching models for N-body trajectory prediction still have gaps in prior distribution design. They typically adopt trivial or simple, observation-agnostic priors such as isotropic Gaussian,

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which overlook two vital properties of N-body dynamics. The first is temporal correlations that the future trajectories exhibit inherent continuity, inertia, and temporal dependencies with past states. The second is spatial symmetries, as N-body trajectories are generally equivariant under global rotations and translations.

Thus, we propose GP-EquiFlow, an SE(3)-equivariant flow matching framework integrated with vector-valued Gaussian processes (VVGPs) (Alvarez et al., 2012) for N-body trajectory prediction. By leveraging VVGPs, we construct a data-dependent prior distribution that aligns with the temporal correlation structure of the observed trajectories. Meanwhile, we incorporate equivariance into both the prior distribution and the time-dependent vector field modeling, ensuring the entire framework is SE(3)-equivariant and respects the fundamental symmetries of N-body systems. In summary, our main contributions are as follows:

- We propose GP-EquiFlow, a unified flow matching framework that models an SE(3)-equivariant distribution of N-body future trajectories conditioned on historical observations. To capture spatial-temporal features and preserve symmetries, GP-EquiFlow leverages equivariant message-passing graph neural networks and the temporal cross-attention to effectively learn the ground-truth vector field.
- We design a data-dependent prior distribution based on vector-valued Gaussian processes. The proposed prior distribution not only sufficiently utilizes the temporal correlations in observed trajectories but also ensures SE(3)-equivariance, thereby naturally coupling the prior with the data distribution.
- Extensive experiments on benchmarks of N-body physical simulations and molecular dynamics demonstrate that GP-EquiFlow outperforms state-of-the-art methods in prediction accuracy while requiring fewer sampling steps.

2. Preliminaries

Geometric Trajectories of N-Body Systems To describe geometric trajectories of N-Body systems, we first define the *geometric trajectory graph* as $\mathcal{G} := (\mathbf{X}, \mathbf{H}, \mathcal{E})$, where $\mathbf{X} := [\mathbf{x}^0, \dots, \mathbf{x}^{T-1}] \in \mathbb{R}^{N \times 3 \times T}$ denotes the 3D position trajectories of all N particles over T timesteps, $\mathbf{x}^t \in \mathbb{R}^{N \times 3}$ is the particle state at timestep t , $\mathbf{H} \in \mathbb{R}^{N \times C_h \times T}$ represents the particle features of dimension C_h , and $\mathcal{E}$ encodes the system topology governing inter-particle interactions.

Invariance and Equivariance For N-body systems, the underlying dynamics are typically equivariant under global rotations and translations (Satorras et al., 2021). In this

work, we focus on the actions of the special Euclidean group SE(3) with elements $g = (\mathbf{R}, \mathbf{t}) \in \text{SE}(3)$, where $\mathbf{R} \in \text{SO}(3)$ is the rotation matrix and $\mathbf{t} \in \mathbb{R}^3$ is the translation vector. The left group actions of SE(3) on a particle position $\mathbf{x} \in \mathbb{R}^3$ is given by $L_g \mathbf{x} = \mathbf{R}\mathbf{x} + \mathbf{t}$. For mappings, let M and N be the input and target spaces acted on by SE(3) via left group actions $L_g : M \rightarrow M$ and $S_g : N \rightarrow N$. A smooth map $\phi : M \rightarrow N$ is SE(3)-invariant if its image remains unchanged under SE(3) group actions, i.e., $\phi(L_g \mathbf{x}) = \phi(\mathbf{x})$, while ϕ is SE(3)-equivariant if it commutes with SE(3) actions on the input and target space, i.e., $\phi(L_g \mathbf{x}) = S_g \phi(\mathbf{x})$. Furthermore, a conditional distribution $p(\mathbf{x}|\mathbf{y})$ is SE(3)-equivariant if SE(3) group acting on $\mathbf{y}$ synchronously acts on $\mathbf{x}$, i.e., $p(L_g \mathbf{x} | S_g \mathbf{y}) = p(\mathbf{x} | \mathbf{y})$.

Flow Matching Flow Matching (Lipman et al., 2023) is an efficient ODE-based generative framework that aims to learn a time-dependent vector field $u_\tau : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ which induces a probability path p_τ connecting an initial distribution p_0 and a target distribution p_1 . Specifically, for a target data point $x_1 \sim p_1$, we can construct a *conditional probability path* $p_\tau(\cdot|x_1)$ such that $p_0(\cdot|x_1) = p_0$ and $p_1(\cdot|x_1) = \delta_{x_1}$, i.e., the Dirac distribution at x_1 . Typically, samples from the conditional probability path $x_\tau \sim p_\tau(\cdot|x_1)$ are formulated as $x_\tau = a_\tau x_1 + b_\tau x_0$, where $x_0 \sim p_0$, and a_τ and b_τ are differentiable and monotonic functions with $a_0 = 0, b_0 = 1$ and $a_1 = 1, b_1 = 0$ ensuring $p_\tau(\cdot|x_1)$ starts from p_0 and terminates at δ_{x_1} . The ground-truth *conditional vector field* $u_\tau(x_\tau|x_1)$ governing the evolution of the conditional probability path is defined via the ODE $\dot{x}_\tau = u_\tau(x_\tau|x_1)$, which can be derived analytically as

$$u_\tau(x_\tau|x_1) = \dot{a}_\tau x_1 + \dot{b}_\tau x_0, \quad (1)$$

where $\dot{a}_\tau = \partial_\tau a_\tau$ and $\dot{b}_\tau = \partial_\tau b_\tau$ denote the time derivatives of a_τ and b_τ , respectively. The training objective of conditional flow matching (CFM) is to minimize the mean square error (MSE) between a parameterized vector field $u_\tau^\theta(x_\tau)$ and the ground-truth conditional vector field $u_\tau(x_\tau|x_1)$, i.e.,

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{\tau, x_1, x_\tau} \|u_\tau^\theta(x_\tau) - u_\tau(x_\tau|x_1)\|^2, \quad (2)$$

where $\tau \sim \text{U}(0, 1)$, $x_1 \sim p_1$, and $x_\tau \sim p_\tau(\cdot|x_1)$. Notably, minimizing this loss is equivalent to minimizing the loss between $u_\tau^\theta(x_\tau)$ and marginal vector field $u_\tau(x_\tau)$ that drives the marginal probability path p_τ from p_0 to target distribution p_1 . During inference, samples from p_1 can be obtained by integrating the ODE $\dot{x}_\tau = u_\tau^\theta(x_\tau)$ over $\tau \in [0, 1]$, initialized from samples $x_0 \sim p_0$.

3. Methodology

3.1. Problem Formulation

Consider a geometric trajectory graph $\mathcal{G} = (\mathbf{X}, \mathbf{H}, \mathcal{E})$ defined in Section 2. For the trajectory $\mathbf{X}$, we can split it

into two disjoint segments as $\mathbf{X} = (\mathbf{X}^{\text{obs}}, \mathbf{X}^{\text{pred}})$, where $\mathbf{X}^{\text{obs}} = [\mathbf{x}^0, \dots, \mathbf{x}^{T_{\text{obs}}-1}] \in \mathbb{R}^{N \times 3 \times T_{\text{obs}}}$ denotes the fully observed trajectory over T_{obs} timesteps and $\mathbf{X}^{\text{pred}} = [\mathbf{x}^{T_{\text{obs}}}, \dots, \mathbf{x}^{T-1}] \in \mathbb{R}^{N \times 3 \times T_{\text{pred}}}$ with $T_{\text{pred}} = T - T_{\text{obs}}$ denotes the future trajectories to be predicted. Our goal is to learn a generative model $p_\theta(\mathbf{X}^{\text{pred}}|\mathbf{X}^{\text{obs}}, \mathbf{H}, \mathcal{E})$ that approximates the ground-truth conditional distribution $q(\mathbf{X}^{\text{pred}}|\mathbf{X}^{\text{obs}}, \mathbf{H}, \mathcal{E})$. Moreover, since the evolution dynamics of N-body systems are typically symmetric, our model should satisfy SE(3)-equivariance, i.e., if the observed trajectory is rotated/translated, the predicted trajectories sampled from p_θ should also be rotated/translated,

$$p_\theta(\mathbf{X}^{\text{pred}}|\mathbf{X}^{\text{obs}}, \mathbf{H}, \mathcal{E}) = p_\theta(g \cdot \mathbf{X}^{\text{pred}}|g \cdot \mathbf{X}^{\text{obs}}, \mathbf{H}, \mathcal{E}), \quad (3)$$

for all $g \in \text{SE}(3)$. Here, we assume that the node features and the system topology $(\mathbf{H}, \mathcal{E})$ are time-invariant and SE(3)-invariant. Accordingly, we omit $(\mathbf{H}, \mathcal{E})$ in subsequent discussions for simplicity, focusing exclusively on trajectory modeling. The conditional generative model $p_\theta(\mathbf{X}^{\text{pred}}|\mathbf{X}^{\text{obs}})$ can be decomposed into an SE(3)-equivariant conditional prior $q_0(\mathbf{X}_0|\mathbf{X}^{\text{obs}})$ and an SE(3)-equivariant generative process $p_\theta(\mathbf{X}^{\text{pred}}|\mathbf{X}_0)$,

$$p_\theta(\mathbf{X}^{\text{pred}}|\mathbf{X}^{\text{obs}}) = \int p_\theta(\mathbf{X}^{\text{pred}}|\mathbf{X}_0)q_0(\mathbf{X}_0|\mathbf{X}^{\text{obs}})d\mathbf{X}_0. \quad (4)$$

Notably, constructing the prior q_0 using a non-equivariant standard Gaussian distribution $\mathcal{N}(0, \mathbf{I})$ or a fixed center-of-mass (CoM) Gaussian distribution $\mathcal{N}(\text{CoM}(\mathbf{x}^{T_{\text{obs}}-1}), \mathbf{I})$ fails to fully exploit the structural information encoded in the observed trajectory. Thus, in the next section, we demonstrate how to model the conditional prior and generative process while capturing both the temporal and spatial features of the N-body trajectory.

3.2. Equivariant Data-Dependent Prior Distribution

Prior work (Albergo et al., 2024; Tong et al., 2024) has shown that constructing a data-dependent prior distribution p_0 close to the data distribution p_1 can effectively lower the transport cost, straighten the learned vector field, and improve the model performance. Given observed trajectories, vector-valued Gaussian processes are well-suited for modeling this data-dependent prior as they naturally capture temporal correlations and accommodate the multi-output structure of trajectories. For each particle i in the N-body system, the 3D position trajectories can be modeled via VVGPs with time $\mathbf{T}$ as the input and observed trajectories as conditions. For a prediction timestep $t \in \mathbf{T}^{\text{pred}}$, the position $\mathbf{x}_i^t \in \mathbb{R}^3$ of particle i follows a Gaussian form:

$$p(\mathbf{x}_i^t|\mathbf{T}^{\text{obs}}, \mathbf{X}_i^{\text{obs}}, t) = \mathcal{N}(\mu(t), \Sigma(t)), \quad (5)$$

where $\mathbf{X}_i^{\text{obs}}$ denotes the observed trajectories of particle i , and the mean function $\mu(t)$ and covariance function $\Sigma(t)$ are given by

$$\mu(t) = \mathbf{K}(t, \mathbf{T}^{\text{obs}}).$$

$$(\mathbf{K}(\mathbf{T}^{\text{obs}}, \mathbf{T}^{\text{obs}}) + \Sigma \otimes \mathbf{I}_{T_{\text{obs}}})^{-1} \bar{\mathbf{X}}^{\text{obs}}, \quad (6)$$

$$\Sigma(t) = \mathbf{K}(t, t) - \mathbf{K}(t, \mathbf{T}^{\text{obs}}).$$

$$(\mathbf{K}(\mathbf{T}^{\text{obs}}, \mathbf{T}^{\text{obs}}) + \Sigma \otimes \mathbf{I}_{T_{\text{obs}}})^{-1} \mathbf{K}(t, \mathbf{T}^{\text{obs}})^\top. \quad (7)$$

Here, $\mathbf{K}(t, \mathbf{T}^{\text{obs}})$ is the cross kernel quantifying the correlation between the prediction timestep and the observed timesteps, while $\mathbf{K}(\mathbf{T}^{\text{obs}}, \mathbf{T}^{\text{obs}})$ captures correlations among observed timesteps. Σ denotes observation noise, and $\bar{\mathbf{X}}^{\text{obs}} \in \mathbb{R}^{3T_{\text{obs}}}$ is the flattened vector of observed trajectory data.

A key design for VVGPs is the construction of valid matrix-valued covariance kernels $\mathbf{K}$. The Linear Model of Coregionalization (LMC) is a widely adopted framework for constructing such kernels (Goovaerts, 1997). Specifically, for any two timesteps t_i, t_j , the covariance kernel $\mathbf{K}(t_i, t_j)$ decomposes into a sum of separable kernels as

$$\mathbf{K}(t_i, t_j) = \sum_{q=1}^Q \mathbf{B}_q k_q(t_i, t_j), \quad (8)$$

where $\mathbf{B}_q \in \mathbb{R}^{3 \times 3}$ denotes the *coregionalization matrix* modeling correlations across the three Cartesian dimensions of particle trajectory, and $k_q(t_i, t_j)$ is the time-dependent kernel modeling temporal correlations. To capture the diverse temporal dynamics of 3D trajectories, we employ three kernel types: the radial basis function (RBF) kernel, linear (LI) kernel, and periodic (PE) kernel, defined as

$$\begin{aligned} k_{\text{RBF}}(t_i, t_j) &= \sigma_r^2 \exp(-\frac{(\delta t)^2}{2l_r^2}), & k_{\text{LI}}(t_i, t_j) &= \sigma_l^2 t_i t_j, \\ k_{\text{PE}}(t_i, t_j) &= \sigma_p^2 \exp(-\frac{2 \sin^2(\pi |\delta t|/T_p)}{l_p^2}), \end{aligned} \quad (9)$$

where $\delta t = t_i - t_j$, $\sigma_r, \sigma_l, \sigma_p$ denote the amplitude hyperparameters that scale the magnitude, the length-scale hyperparameters l_r, l_p are positive scalars that determine the decay rate, and T_p is the period parameter that characterizes the repeating patterns in trajectories. The RBF kernel is designed to fit smooth trajectories, as it efficiently captures continuous, non-abrupt dynamic patterns in trajectory data. The LI kernel captures linear trends within trajectories, while the PE kernel is specifically designed to model repeating patterns by tuning the period parameter T_p .

Next, to enforce SE(3)-equivariance for the conditional prior distribution $p_0(\mathbf{X}_0|\mathbf{X}^{\text{obs}})$, we impose two additional constraints on the covariance kernel. First, the coregionalization matrix is fixed to the identity matrix as $\mathbf{B}_q = \mathbf{I}_3$. Then, the observation noise is assumed to be isotropic with $\Sigma = \sigma^2 \mathbf{I}_3$. Under these constraints, the following theorem establishes the SO(3)-equivariance of the proposed prior.

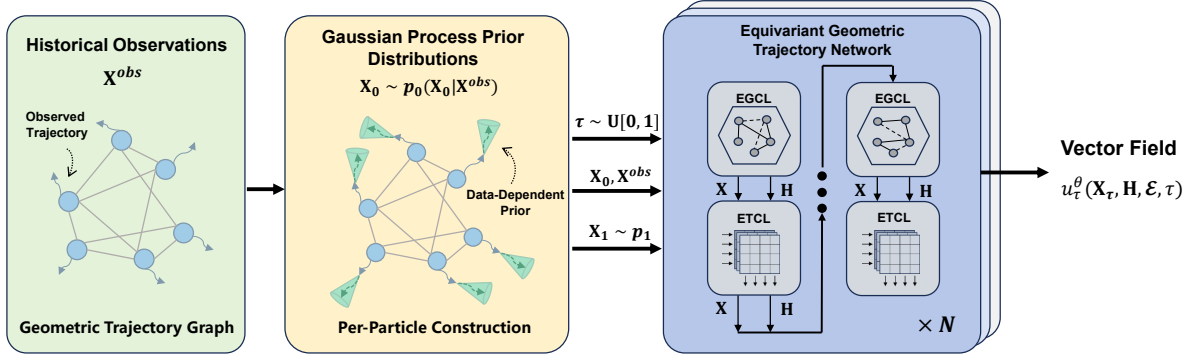


Figure 1. Overview of GP-EquiFlow. Given historical observation trajectories $\mathbf{X}^{\text{obs}}$, we construct an observation-dependent prior $\mathbf{X}_0$ via vector-valued Gaussian processes. For each particle, the solid blue line denotes its observed trajectory, the dashed green line represents the mean predicted by the VVGP, and the green cone indicates the predictive covariance. Then, we utilize a stack of repeating equivariant graph convolution layers (EGCL) and equivariant temporal cross-attention layers (ETCL) to learn the vector field $u_\tau(\mathbf{X}_\tau)$.

Theorem 3.1. Let η denote the optimized kernel hyperparameters learned from the observations $\{\mathbf{T}^{\text{obs}}, \mathbf{X}^{\text{obs}}\}$, with the coregionalization matrix satisfying $\mathbf{B}_q = \mathbf{I}_3$. Assume observation noise is isotropic, i.e., $\Sigma = \sigma^2 \mathbf{I}_3$. For any $g \in \text{SO}(3)$, $t \in \mathbf{T}^{\text{pred}}$, and rotated observations $\{\mathbf{T}^{\text{obs}}, g \cdot \mathbf{X}^{\text{obs}}\}$, the rotational equivariance of the conditional prior distribution holds as

$$p(\mathbf{x}_i^t | \mathbf{T}^{\text{obs}}, \mathbf{X}_i^{\text{obs}}; \eta) = p(g \cdot \mathbf{x}_i^t | \mathbf{T}^{\text{obs}}, g \cdot \mathbf{X}_i^{\text{obs}}; \eta) \quad (10)$$

Remark 3.1. The proof is detailed in Appendix B. The introduced constraints imply that the (x, y, z) dimensions of particle trajectories are isotropic in prediction, meaning the kernel’s covariance structure is shared uniformly across all spatial dimensions. This aligns with the dynamics of N-body systems in the absence of anisotropic external fields, which exhibit no directional preference.

3.3. Training and Inference

As outlined in Section 2, conditional flow matching requires artificial construction of a conditional probability flow path $x_\tau \sim p_\tau(\cdot | x_1)$ and a corresponding conditional vector field $u_\tau(x_\tau | x_1)$. Here, we adopt the widely used Gaussian probability path with linear interpolation coefficients $a_\tau = \tau$, $b_\tau = 1 - \tau$, which yields

$$\mathbf{X}_\tau = \tau \mathbf{X}_1 + (1 - \tau) \mathbf{X}_0, u_\tau(\mathbf{X}_\tau | \mathbf{X}_1) = \mathbf{X}_1 - \mathbf{X}_0, \quad (11)$$

where $\mathbf{X}_0 \sim p_0(\mathbf{X}_0 | \mathbf{X}^{\text{obs}})$ is the data-dependent prior in CFM constructed via VVGPs and $\mathbf{X}_1$ denotes the ground-truth future 3D position trajectories. The CFM training loss

is then given by

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{\tau, \mathbf{X}_1, \mathbf{X}_0} \|\mathbf{u}_\tau^\theta(\mathbf{X}_\tau, \mathbf{H}, \mathcal{E}, \tau) - (\mathbf{X}_1 - \mathbf{X}_0)\|^2. \quad (12)$$

In practice, we adopt a two-stage training pipeline to first train VVGPs for constructing the equivariant prior distribution, then optimize the CFM loss over the constructed priors. Figure 1 provides the overview of GP-EquiFlow.

Training For each particle in the N-body system, we independently train a dedicated VVGP on its observed trajectories. This particle-wise modeling preserves the structural independence of individual particle dynamics while enabling efficient parallel training. We first normalize the observed timesteps $\mathbf{T}^{\text{obs}}$ and the particle trajectory $\mathbf{X}_i^{\text{obs}}$ to zero mean and unit variance to stabilize VVGP optimization and explicitly enable translational equivariance.

$$\mathbf{T}_{\text{norm}}^{\text{obs}} = \frac{\mathbf{T}^{\text{obs}} - \mu_{\mathbf{T}}^{\text{obs}}}{\sigma_{\mathbf{T}}^{\text{obs}}}, \quad \mathbf{X}_{i,\text{norm}}^{\text{obs}} = \frac{\mathbf{X}_i^{\text{obs}} - \mu_{\mathbf{X}_i^{\text{obs}}}^{\text{obs}}}{\sigma_{\mathbf{X}_i^{\text{obs}}}^{\text{obs}}}, \quad (13)$$

where $\mu_{\mathbf{T}}^{\text{obs}}, \sigma_{\mathbf{T}}^{\text{obs}} \in \mathbb{R}$ are the mean and standard deviation of the observed timesteps, and $\mu_{\mathbf{X}_i^{\text{obs}}}^{\text{obs}}, \sigma_{\mathbf{X}_i^{\text{obs}}}^{\text{obs}} \in \mathbb{R}^3$ are the dimension-wise mean and standard deviation of the observed 3D position trajectories for particle i . We then select a suitable kernel combination and optimize the kernel hyperparameters η by minimizing the negative log-likelihood (NLL) loss. To balance the accurate prior fitting and training efficiency, we optimize the NLL utilizing the L-BFGS-B optimizer with only 5 training epochs per particle. With the optimized hyperparameters, we compute the normalized predictive mean $\mu_{\text{norm}}(t) \in \mathbb{R}^3$ and normalized predictive covariance matrix $\Sigma_{\text{norm}}(t) \in \mathbb{R}^{3 \times 3}$ for future timesteps $t \in \mathbf{T}^{\text{pred}}$ via Eqs. (6) and (7). Next, we denormalize the

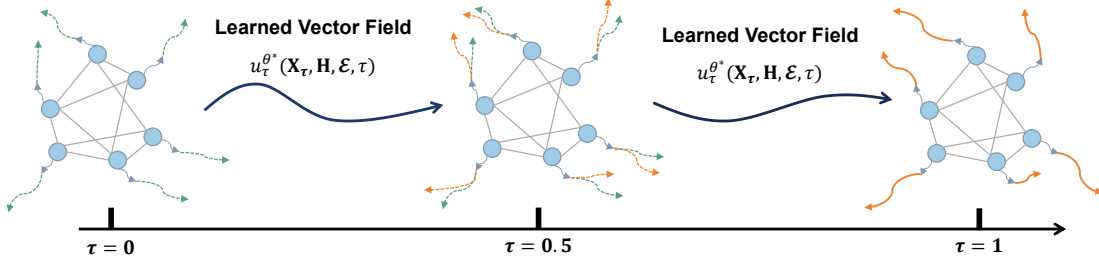


Figure 2. At $\tau = 0$, we sample prior predicted trajectories (green dashed lines) from the observation-dependent prior distribution. Driven by the learned vector field $u_\tau^{\theta^*}(\cdot)$, they evolve toward the target distribution (orange dashed lines). At $\tau = 1$, GP-EquiFlow generates the final predicted trajectories (solid orange lines).

predicted mean $\mu(t)$ to recover the original scale of the particle trajectory data.

$$\mu(t) = \mu_{\text{norm}}(t) \odot \sigma_{\mathbf{X}_i^{\text{obs}}} + \mu_{\mathbf{X}_i^{\text{obs}}}, \quad (14)$$

where $\odot$ denotes Hadamard product. It is evident that the normalization-denormalization procedure provably enforces translational equivariance of the predicted particle trajectories. Formally, for any constant translation $\mathbf{t} \in \mathbb{R}^3$ and prediction timestep $t \in \mathbf{T}^{\text{pred}}$, we have

$$\mu(t, \mathbf{X}_i^{\text{obs}} + \mathbf{t}) = \mu(t, \mathbf{X}_i^{\text{obs}}) + \mathbf{t}. \quad (15)$$

Notably, the normalized predictive covariance matrix $\Sigma(t)$ already encodes the temporal structure with its scale growing as the prediction timestep moves further from the observed data. Direct denormalization of $\Sigma_{\text{norm}}(t)$ would break the rotational invariance of the covariance structure. To resolve this, we introduce a scaling hyperparameter $w > 0$ to calibrate the scale of the predictive covariance matrix across different scenarios:

$$\Sigma(t) = w \cdot \Sigma_{\text{norm}}(t). \quad (16)$$

Combining the SO(3)-equivariance in Theorem 3.1 with the translational equivariance, we thus construct an SE(3)-equivariant data-dependent prior distribution $p_0(\mathbf{X}_0|\mathbf{X}^{\text{obs}})$. During CFM training, we replace the observed segments of $\mathbf{X}_0$ with ground-truth $\mathbf{X}^{\text{obs}}$ and provide a binary mask to flag observed timesteps. At the model output, we retain only the trajectory segments requiring prediction instead of all timesteps, ensuring efficiency and alignment with our objectives. Our method can be generalized to arbitrary observation-based N-body trajectory prediction tasks, including sparse discrete observation sequences, endpoint-conditioned simulation, and trajectory imputation.

Model Architecture The backbone of our method is the Equivariant Geometric Trajectory Network (ETGN) (Han

et al., 2024), composed of Equivariant Graph Convolution Layers (EGCL) and Equivariant Temporal Cross-Attention Layers (ETCL). We adapt ETGN to flow matching to learn the target vector field $u_\tau(\cdot)$. Since ETGN is SO(3)-equivariant and translationally invariant, combining it with the proposed SE(3)-equivariant prior p_0 enables the entire flow matching framework to satisfy SE(3)-equivariance.

Theorem 3.2. *Let $p_\tau(\mathbf{X}_\tau|\mathbf{X}^{\text{obs}})$ denote the marginal probability path. If $p_0(\mathbf{X}_0|\mathbf{X}^{\text{obs}})$ is SE(3)-equivariant and $u_\tau^\theta(\mathbf{X}_\tau)$ is SO(3)-equivariant and translationally invariant, the probability path $p_\tau(\mathbf{X}_\tau|\mathbf{X}^{\text{obs}})$ is SE(3)-equivariant for all $\tau \in [0, 1]$.*

Remark 3.2. The detailed proof is provided in Appendix B. The Theorem 3.2 establishes that if the constructed data-dependent prior $p_0(\mathbf{X}_0|\mathbf{X}^{\text{obs}})$ satisfies SE(3)-equivariance, and the backbone (ETGN) learning the vector field $u_\tau^\theta(\mathbf{X}_\tau)$ is SO(3)-equivariant and translationally invariant, the entire probability path $p_\tau(\mathbf{X}_\tau|\mathbf{X}^{\text{obs}})$ is SE(3)-equivariant. This implies that for observed trajectories under SE(3) group actions, GP-EquiFlow can directly generate predicted trajectories under the same group actions without retraining.

Inference We first train the VVGPs to derive the data-dependent prior distribution p_0 , and sample prior trajectories from it $\mathbf{X}_0 \sim p_0$. Then, we forward simulate the ODE $\dot{\mathbf{X}}_\tau = u_\tau^\theta(\mathbf{X}_\tau)$ from $\tau = 0$ to $\tau = 1$ with the learned vector field $u_\tau^\theta(\mathbf{X}_\tau)$ via the Euler method to generate predicted trajectories $\mathbf{X}^{\text{pred}}$. For trajectory prediction under SE(3) group actions, we directly employ the previously optimized hyperparameters η^* instead of retraining VVGPs, preserving the SE(3)-equivariance of the prior distribution p_0 while avoiding additional computational overhead.

4. Related Work

Research on learning-based geometric trajectory simulation focuses on designing models that capture temporal correlations, model inter-particle interactions and ensure spatial symmetries. To this end, equivariant graph neural networks such as EGNN (Satorras et al., 2021), GMN (Huang et al., 2022), SEGNN (Brandstetter et al., 2022), CEGNN (Ruhe et al., 2023) and PONITA (Bekkers et al., 2024) leverage SE(3)/E(3)-equivariance to align with the symmetries of geometric N-body systems (Batzner et al., 2022; Cen et al., 2024). These methods adopt a frame-to-frame prediction paradigm and tend to accumulate errors during long-term forecasting. To mitigate this issue, some approaches integrate ordinary differential equations (Liu et al., 2024), Newton-Cotes formulas (Guo et al., 2023) and neural operators (Xu et al., 2024) while others adopt sequence-to-sequence prediction frameworks (Xu et al., 2023a). In contrast, recent generative models (Wen et al., 2023; Han et al., 2024; ten Brinke et al., 2025) have achieved success by effectively modeling the stochasticity inherent in real dynamic systems and enabling direct trajectory generation. Our work focuses on this domain and aims to design a prior distribution that effectively captures dynamic patterns from historical observations while satisfying symmetry constraints. This design simplifies the model’s learning of the mapping from the prior to the data distribution. Additionally, numerous flow matching and diffusion model-based methods focus on geometric data generation including molecular generation (Xu et al., 2025a; Hassan et al., 2024; Xu et al., 2022a; 2023b; Bergues et al., 2026) and protein design (Watson et al., 2023; Yim et al., 2023; Yue et al., 2025). However, these methods primarily generate single-frame snapshots of geometric N-body systems. Our work, by contrast, focuses on the temporal evolution of such systems. Furthermore, generative models have also demonstrated effectiveness in time series forecasting (Shen & Kwok, 2023; Kolloviev et al., 2025; 2023; Xu et al., 2025b). Our method draws inspiration from these works and extends them to N-body systems with explicit symmetry considerations.

5. Experiments

We evaluate the proposed GP-EquiFlow on N-body system simulation and molecular dynamics for trajectory forecasting tasks. We primarily focus on the impact of the prior distribution on the performance of generative models under conditional settings, and thus strictly follow the experimental setups and model details in (Han et al., 2024) to ensure fair comparison. Additionally, ablation studies on different prior distributions are detailed in Section 5.4, and the impact of the sampling steps during inference is presented in Section 5.3. The standard deviations of experimental results, training hyperparameters, and compute resources can be

found in Appendix C.

5.1. N-body Systems

Table 1. Average Displacement Error (ADE) and Final Displacement Error (FDE) for trajectory prediction over Gravity, Spring, and Charged datasets, averaged over 5 runs. The best results are **bolded** and the second best are underlined.

Dataset	Gravity		Spring		Charged	
	ADE	FDE	ADE	FDE	ADE	FDE
TFN	0.327	0.761	0.1013	0.2364	0.330	0.754
SE(3)-Tr	0.338	0.830	0.0865	0.2043	0.395	0.936
EGNN	0.310	0.709	0.0101	0.0231	0.186	0.426
SVAE	0.582	1.101	0.0120	0.0209	0.378	0.732
EqMotion	0.302	0.671	0.0134	0.0358	0.141	0.310
GeoTDM	<u>0.256</u>	<u>0.613</u>	<u>0.0030</u>	<u>0.0079</u>	<u>0.110</u>	<u>0.258</u>
GP-EquiFlow	0.105	0.274	0.0019	0.0039	0.068	0.152

Datasets and Setup Following GeoTDM (Han et al., 2024), we evaluate our method on three N-body simulation benchmarks. **1. Charged Particles** ($N = 5$), where particles with random charges ($+1/-1$) interact via Coulomb forces; **2. Spring Dynamics** ($N = 5$), where particles are connected by springs with probability 0.5, governed by Hooke’s law; **3. Gravity System** ($N = 10$), where particles with random masses and initial velocities driven by gravitational dynamics. For all datasets, each trajectory consists of $T = 30$ timesteps, with the first $T_{\text{obs}} = 10$ timesteps utilized as the conditioning observations to predict the subsequent 20 timesteps. All datasets are split into 3,000 trajectories for training, 2,000 for validation, and 2,000 for testing.

Baselines We compare our GP-EquiFlow against representative baselines across three categories of trajectory forecasting methods. Frame-to-frame prediction models: Tensor Field Networks (Thomas et al., 2018), SE(3)-Transformers (Fuchs et al., 2020), and EGNN (Satorras et al., 2021); Deterministic trajectory forecasting models: EqMotion (Xu et al., 2023a); and Probabilistic models: SVAE (Xu et al., 2022b) and GeoTDM (Han et al., 2024).

Metrics Following standard practice in trajectory forecasting, we adopt Average Displacement Error (ADE) and Final Displacement Error (FDE) as the evaluation metrics. ADE quantifies the mean Euclidean distance between predicted and ground-truth positions over the entire prediction window, defined as $\text{ADE} = \frac{1}{N(T-T_{\text{obs}})} \sum_{t=T_{\text{obs}}}^{T-1} \sum_{i=1}^N \|\hat{\mathbf{x}}_i^{(t)} - \mathbf{x}_i^{(t)}\|_2$, FDE measures the drift at the final prediction timestep, $\text{FDE} = \frac{1}{N} \sum_{i=1}^N \|\hat{\mathbf{x}}_i^{(T-1)} - \mathbf{x}_i^{(T-1)}\|_2$, where $\hat{\mathbf{x}}_i^{(t)}$ denotes the predicted position of particle i at timestep t , and $\mathbf{x}_i^{(t)}$ is the corresponding ground-truth position.

Implementation We represent the input data as fully connected geometric graphs $\mathcal{G} = (\mathbf{X}, \mathbf{H}, \mathcal{E})$, where node features $\mathbf{H}$ and the system topology $\mathcal{E}$ are assumed to be time-invariant. In detail, each node is associated with a one-dimensional feature, for instance, in the *Charged Particles*, the node feature corresponds to the particle charge (+1/−1). The kernel of the Gaussian processes is a sum of the RBF and linear (LI) kernels, with the scaling hyperparameter set to $w = 1$. The number of sampling steps used during inference is set to 10 for all experiments. Detailed discussions on the choice of kernel function and scaling hyperparameter are provided in Appendix C.

Results The quantitative results in Table 1 demonstrate the consistent superiority of our GP-EquiFlow across all three N-body system datasets. Specifically, our method outperforms the previous state-of-the-art probabilistic model, GeoTDM, reducing the ADE by 44.6% and FDE by 48.9% on average across all three datasets. In contrast to GeoTDM that models the prior via an additional ETGN module, our GP-based prior construction yields a more reliable and interpretable prior distribution that is closer to the target distribution. This design not only improves prediction accuracy but also incurs lower computational overhead. Furthermore, the superior FDE performance of GP-EquiFlow highlights its capacity to effectively mitigate prediction errors for longer-horizon forecasting.

5.2. Molecular Dynamics

Datasets and Setup We further evaluate GP-EquiFlow on the MD17 benchmark, which consists of molecular dynamics trajectories for 8 molecules generated via density functional theory (DFT) simulations. Consistent with (Han et al., 2024), we retain hydrogen atoms as their trajectories exhibit more significant vibrations, which elevates the prediction challenge. The molecules vary in size from 9 atoms (Ethanol and Malonaldehyde) to 21 atoms (Aspirin). Node features $\mathbf{H}$ are one-hot encodings of atomic categories, and the system topology $\mathcal{E}$ is constructed by connecting nodes within three-hops connections based on atomic bonds. For each molecule, we downsample trajectories by keeping every 10-th timestep. We adopt a temporal split of 60%, 20%, and 20% for training, validation, and testing, respectively, and sample 5000, 1000, 1000 trajectories from each partition to construct the corresponding datasets. The kernel of the Gaussian processes is a sum of the RBF and PE kernels, with the scaling hyperparameter set to $w = 1$. The prediction task involves a horizon of $T = 30$ frames, conditioned on the first $T_{\text{obs}} = 10$ frames. The baselines and metrics are the same as the N-body experiments.

Results As shown in Table 2, GP-EquiFlow achieves the best performance on 6 out of 8 molecules, and ranks top-2

across all datasets. Compared with the previous state-of-the-art method, GP-EquiFlow reduces ADE by 8.8% and FDE by 10.7% on average across all molecules. This confirms that for molecular dynamics with significant vibrations, in contrast to the relatively smooth trajectories of N-body systems, the proposed GP-based prior can still capture trajectory patterns from historical observations. Notably, the performance improvements of GP-EquiFlow on MD17 are less pronounced than those on N-body simulations. This discrepancy may arise from two potential characteristics of molecular dynamics. First, molecular dynamics tend to operate within a smaller spatial range compared with N-body system dynamics. Second, significant vibrations are likely to complicate the characterization of trajectory temporal correlations using only observations. These two factors may collectively result in greater entanglement between the constructed observation-dependent prior and the target distribution. Consequently, the performance gap between our GP-based prior and GeoTDM’s ETGN-based prior is narrower than that observed in N-body experiments.

5.3. Studies on Different Sampling Steps

We further conduct experiments on the *Spring* ($N = 5$), *Gravity* ($N = 10$), *Benzene* ($N = 12$), and *Toluene* ($N = 15$) datasets to investigate the impact of sampling steps during inference. The results are presented in Table 3, where the **Prior** column reports the error between trajectories sampled directly from our GP-based priors and the ground-truth trajectories. GP-EquiFlow achieves accurate prediction even with only 2 steps and reaches the best prediction accuracy on the *Gravity*, *Benzene*, and *Toluene* datasets with 10 steps. This verifies that our GP-based prior is of high quality and well-aligned with the target distribution, enabling the generated trajectories to flow from the prior to the target distribution for accurate prediction with few steps. Interestingly, increasing the number of sampling steps beyond 10 does not consistently improve performance. On the *Benzene* dataset, the 100-step Euler integration fails to generate converged results, and similar performance degradation is observed on the *Gravity* and *Toluene* datasets when sampling steps exceed 10. This can be attributed to the singularity at $\tau = 1$ as noted in (Lipman et al., 2024). Excessive sampling steps tend to amplify the impact of such unstable vector fields, which in turn accumulates numerical errors and ultimately degrades prediction performance or even causes divergence in extreme cases.

5.4. Ablation Studies on Different Priors

We conduct ablation studies to investigate the impact of different priors p_0 on trajectory prediction tasks. Here, we additionally consider five distinct priors for comparison, including the standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$, the Gaussian distribution centered at the last observed posi-

Table 2. Average Displacement Error (ADE) and Final Displacement Error (FDE) for trajectory prediction on MD17, averaged over 5 runs. The best results are **bolded** and the second best are underlined.

Molecule	Aspirin		Benzene		Ethanol		Malonaldehyde		Naphthalene		Salicylic		Toluene		Uracil	
Metric	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE
RF	0.303	0.442	0.120	0.194	0.374	0.515	0.297	0.454	0.168	0.185	0.261	0.343	0.199	0.249	0.239	0.272
TFN	0.133	0.268	0.024	0.049	0.201	0.414	0.184	0.386	0.072	0.098	0.115	0.223	0.090	0.150	0.090	0.159
SE(3)-Tr.	0.294	0.556	0.027	0.056	0.188	0.359	0.214	0.456	0.069	0.103	0.189	0.312	0.108	0.184	0.107	0.196
EGNN	0.267	0.564	0.024	0.042	0.268	0.401	0.393	0.958	0.095	0.133	0.159	0.348	0.207	0.294	0.154	0.282
EqMotion	0.185	0.246	0.029	0.043	0.152	0.247	0.155	0.249	0.073	0.092	0.110	0.151	0.097	0.129	0.088	0.116
SVAE	0.301	0.428	0.114	0.133	0.387	0.505	0.287	0.430	0.124	0.135	0.122	0.142	0.145	0.171	0.145	0.156
GeoTDM	<u>0.107</u>	<u>0.193</u>	<u>0.023</u>	<u>0.039</u>	<u>0.115</u>	<u>0.209</u>	0.107	<u>0.176</u>	<u>0.064</u>	<u>0.087</u>	0.083	0.120	<u>0.083</u>	<u>0.121</u>	<u>0.074</u>	<u>0.099</u>
GP-EquiFlow	0.093	0.179	0.020	0.034	0.105	0.182	<u>0.117</u>	0.161	0.063	0.079	<u>0.096</u>	<u>0.126</u>	0.075	0.109	0.065	0.085

Table 3. Average Displacement Error (ADE) and Final Displacement Error (FDE) for trajectory prediction on the *Spring*, *Gravity*, *Benzene*, and *Toluene* datasets with different Euler sampling steps during inference. Results are averaged over 5 independent runs, with “-” denoting invalid NaN values. The best results are **bolded**.

Metric	Prior		Euler-2		Euler-5		Euler-10		Euler-20		Euler-50		Euler-100	
	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE
Spring	0.886	1.799	0.0292	0.0458	0.0033	0.0075	0.0019	0.0039	0.0017	0.0030	0.0014	0.0030	0.0016	0.0036
Gravity	1.296	2.181	0.112	0.280	0.106	0.276	0.105	0.274	0.121	0.307	0.125	0.318	0.129	0.324
Benzene	1.453	1.611	0.057	0.079	0.024	0.038	0.020	0.034	0.022	0.036	0.036	0.041	-	-
Toluene	1.603	1.687	0.111	0.139	0.089	0.123	0.075	0.109	0.085	0.116	0.099	0.141	0.102	0.142

Table 4. Average Displacement Error (ADE) and Final Displacement Error (FDE) for prediction based on different priors over Gravity, Spring, and Uracil datasets. The best results are **bolded**.

Dataset	Gravity		Spring		Uracil	
Metric	ADE	FDE	ADE	FDE	ADE	FDE
$\mathcal{N}(\mathbf{0}, \mathbf{I})$	0.168	0.363	0.0286	0.0348	0.108	0.145
$\mathcal{N}(\mathbf{x}^{T_{\text{obs}}-1}, \mathbf{I})$	0.148	0.353	0.0128	0.0187	0.092	0.114
$\mathcal{N}(\text{CoM}(\mathbf{x}^{T_{\text{obs}}-1}), \mathbf{I})$	0.242	0.557	0.0074	0.0148	0.087	0.108
Random Walk	0.121	0.286	0.0058	0.0161	0.098	0.112
GeoFlow	0.135	0.312	0.0101	0.0231	0.085	0.104
GP-EquiFlow	0.105	0.274	0.0019	0.0039	0.065	0.085

tion $\mathcal{N}(\mathbf{x}^{T_{\text{obs}}-1}, \mathbf{I})$, the fixed center-of-mass (CoM) prior $\mathcal{N}(\text{CoM}(\mathbf{x}^{T_{\text{obs}}-1}), \mathbf{I})$, the Gaussian random walk prior with the scaling parameter $s = 4$ in (ten Brinke et al., 2025), and GeoFlow, which adapts the prior from (Han et al., 2024) to the conditional flow matching framework. We evaluate these priors on the *Gravity* ($N = 10$), *Spring* ($N = 5$), and *Uracil* ($N = 12$) datasets, with quantitative results presented in Table 4. Compared with the standard unconditioned Gaussian prior, observation-conditioned priors consistently yield improved prediction performance, which validates the effectiveness of leveraging historical observations for prior construction. The Gaussian Random Walk prior delivers strong performance on the *Gravity* and *Spring* datasets, where trajectories are relatively smooth. However, the Gaussian Random Walk prior tends to rely on the average dynamical trend across the entire observed trajectory when generating

prior samples, as its update rule is conditioned on cumulative historical observations. This makes it less adaptable to datasets like *Uracil*, in which atomic trajectories exhibit more vibrations and future trajectories may thus diverge from earlier trends. As a result, Gaussian random walk may construct priors that deviate further from the true target distribution than the Gaussian prior centered at the last observed position, which contributes to its relatively weaker performance. While GeoFlow also incorporates SE(3) equivariance and temporal modeling and achieves competitive results, its over-parameterized coupled prior increases the optimization difficulty, as the downstream flow matching needs to learn mappings from complex evolving prior distributions to the target distribution in the training stage. Our non-parametric VVGP-based prior mitigates this issue with far fewer parameters, enabling easier learning while preserving temporal structure and N-body spatial symmetries. Furthermore, we find that when equipped with an appropriate kernel and proper hyperparameters, the fixed GP prior can outperform its trained counterpart by ensuring complete prior consistency throughout training. Detailed results are provided in the Appendix D.

6. Conclusion

We proposed GP-EquiFlow, a novel and efficient SE(3)-equivariant flow matching model for N-body trajectory prediction conditioned on historical observations. Utilizing vector-valued Gaussian processes, we construct a

data-dependent prior distribution that captures the temporal correlations of trajectories while strictly satisfying SE(3)-equivariance, which enables the model to efficiently learn a flow mapping from the prior distribution to the target data distribution for trajectory generation. Extensive empirical results on various benchmarks demonstrate the superior prediction accuracy of GP-EquiFlow. Our method currently adopts a fixed prediction horizon. For long-term trajectory prediction, an available method is to decompose the target horizon into successive small segments and conduct autoregressive prediction over these segments. Additionally, the backbone architecture of our model can be replaced with more expressive geometric deep learning frameworks to further enhance the model’s representational capacity for complex N-body dynamics. Finally, we plan to explore the integration of one-step generation methods (Geng et al., 2026; Song et al., 2023) into the GP-EquiFlow framework, which is expected to further accelerate sampling and improve prediction accuracy.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Gaussian Processes for Vector-valued Functions

Gaussian Processes (GPs) offer a flexible probabilistic framework for modeling uncertainty in function estimation, and their extension to vector-valued functions facilitates effective handling of multi-output learning tasks. In such scenarios, the objective is to learn a mapping from an input space $\mathcal{X} \subseteq \mathbb{R}^p$ to an output space $\mathbb{R}^D$, where p, D denote the dimensions of input and output spaces, respectively. In contrast to single-output Gaussian processes, vector-valued gaussian processes leverage inherent dependencies between outputs to improve predictive performance.

Formally, a vector-valued function $\mathbf{f} : \mathcal{X} \rightarrow \mathbb{R}^D$ is said to follow a gaussian process if $\mathbf{f} \sim \mathcal{GP}(\mathbf{m}, \mathbf{K})$, where $\mathbf{m} : \mathcal{X} \rightarrow \mathbb{R}^D$ is the mean function with components $m_d(\mathbf{x}) = \mathbb{E}[f_d(\mathbf{x})]$, and $\mathbf{K} : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}^{D \times D}$ is a positive semi-definite matrix-valued covariance kernel. The entries of $\mathbf{K}(\mathbf{x}, \mathbf{x}')$ are given by $\mathbf{K}_{d,d'}(\mathbf{x}, \mathbf{x}') = \text{cov}[f_d(\mathbf{x}), f_{d'}(\mathbf{x}')]$, which quantify the correlation between the d -th and d' -th outputs evaluated at input points $\mathbf{x}, \mathbf{x}'$.

For a set of inputs $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, the prior distribution over the outputs $\mathbf{f}(\mathbf{X}) = [f_1(\mathbf{X})^\top, \dots, f_D(\mathbf{X})^\top]^\top$, where $f_d(\mathbf{X}) = [f_d(\mathbf{x}_1), \dots, f_d(\mathbf{x}_N)]^\top$, follows a multivariate Gaussian distribution

$$\mathbf{f}(\mathbf{X}) \sim \mathcal{N}(\mathbf{0}, \mathbf{K}(\mathbf{X}, \mathbf{X})). \quad (17)$$

Here, without loss of generality, the mean function is assumed to be zero, which can be readily achieved through normalization in practice. The covariance matrix $\mathbf{K}(\mathbf{X}, \mathbf{X}) \in \mathbb{R}^{ND \times ND}$ is a block-partitioned matrix with blocks $\mathbf{K}_{d,d'}(\mathbf{X}, \mathbf{X}) \in \mathbb{R}^{N \times N}$ whose (i, j) -th entry is given by $\mathbf{K}_{d,d'}(\mathbf{x}_i, \mathbf{x}_j)$. For the observed outputs $\mathbf{y} \in \mathbb{R}^D$, the likelihood function is modeled as Gaussian distribution

$$p(\mathbf{y} \mid \mathbf{f}, \mathbf{x}, \Sigma) = \mathcal{N}(\mathbf{f}(\mathbf{x}), \Sigma), \quad (18)$$

where $\Sigma \in \mathbb{R}^{D \times D}$ represents the observation noise variances.

Under the Gaussian likelihood assumption, the predictive distribution for a new input $\mathbf{x}_* \in \mathcal{X}$ can be derived analytically. Given the observations $\{\mathbf{X}, \mathbf{Y}\}$, the posterior distribution of the new output $\mathbf{f}(\mathbf{x}_*) = [f_1(\mathbf{x}_*), \dots, f_D(\mathbf{x}_*)]^\top \in \mathbb{R}^D$ retains the Gaussian form

$$p(\mathbf{f}(\mathbf{x}_*) \mid \mathbf{X}, \mathbf{Y}, \mathbf{x}_*, \theta) = \mathcal{N}(\mu_*(\mathbf{x}_*), \Sigma_*(\mathbf{x}_*)), \quad (19)$$

where θ denotes a set of hyperparameters associated with the matrix-valued covariance kernel $\mathbf{K}(\mathbf{x}, \mathbf{x}')$. The predictive mean $\mu_*(\mathbf{x}_*) \in \mathbb{R}^D$ and predictive covariance $\Sigma_*(\mathbf{x}_*) \in \mathbb{R}^{D \times D}$ are explicitly given by

$$\mu_*(\mathbf{x}_*) = \mathbf{K}(\mathbf{x}_*, \mathbf{X}) (\mathbf{K}(\mathbf{X}, \mathbf{X}) + \Sigma \otimes \mathbf{I}_N)^{-1} \bar{\mathbf{y}}, \quad (20)$$

$$\Sigma_*(\mathbf{x}_*) = \mathbf{K}(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{K}(\mathbf{x}_*, \mathbf{X}) (\mathbf{K}(\mathbf{X}, \mathbf{X}) + \Sigma \otimes \mathbf{I}_N)^{-1} \mathbf{K}(\mathbf{X}, \mathbf{x}_*)^\top. \quad (21)$$

Here, $\mathbf{K}(\mathbf{x}_*, \mathbf{X}) \in \mathbb{R}^{D \times ND}$ is the cross-covariance matrix capturing dependencies between the new output $\mathbf{f}(\mathbf{x}_*)$ and the training outputs $\mathbf{f}(\mathbf{X})$. $\bar{\mathbf{y}} = [y_1^\top, \dots, y_D^\top]^\top \in \mathbb{R}^{ND}$ denotes the vectorized form of the training output $\mathbf{Y}$.

A.1. Linear Model of Coregionalization

The LMC assumes that output components arise from a linear combination of independent latent Gaussian processes, which naturally induces a covariance kernel structured as a sum of separable terms. Formally, each output component $f_d(\mathbf{x})$ can be expressed as

$$f_d(\mathbf{x}) = \sum_{q=1}^Q \sum_{i=1}^{R_q} a_{d,q}^i u_q^i(\mathbf{x}), \quad (22)$$

where $a_{d,q}^i$ are scalar coefficients weighting the contribution of each latent process to the output, Q denotes the number of latent function groups, R_q is the number of independent latent functions within the q -th group, and $u_q^i : \mathcal{X} \rightarrow \mathbb{R}$ are independent latent functions with covariance

$$\text{cov}[u_q^i(\mathbf{x}), u_{q'}^j(\mathbf{x}')] = \begin{cases} k_q(\mathbf{x}, \mathbf{x}') & \text{if } q = q' \text{ and } i = j, \\ 0 & \text{otherwise.} \end{cases} \quad (23)$$

This implies that the latent functions are partitioned into Q groups, Latent functions from different groups are mutually independent, and those within the same group share the same covariance kernel while also being mutually independent.

From this structure, the covariance kernel $\mathbf{K}(\mathbf{x}, \mathbf{x}')$ for the vector-valued function $\mathbf{f}$ can be derived as a sum of separable kernels. Specifically, the cross-covariance between the d -th and d' -th outputs at inputs $\mathbf{x}$ and $\mathbf{x}'$ is

$$\mathbf{K}_{d,d'}(\mathbf{x}, \mathbf{x}') = \sum_{q=1}^Q b_{d,d'}^q k_q(\mathbf{x}, \mathbf{x}'), \quad (24)$$

where $b_{d,d'}^q = \sum_{i=1}^{R_q} a_{d,q}^i a_{d',q}^i$ defines the (d, d') -th entry of $D \times D$ symmetric positive matrix $\mathbf{B}_q$ referred to as the *coregionalization matrix*. Accordingly, the full kernel $\mathbf{K}(\mathbf{x}, \mathbf{x}')$ can be expressed as

$$\mathbf{K}(\mathbf{x}, \mathbf{x}') = \sum_{q=1}^Q \mathbf{B}_q k_q(\mathbf{x}, \mathbf{x}'), \quad (25)$$

B. Theoretical Results and Proofs

Lemma B.1. *Let $p(\mathbf{x} \mid \mathbf{X}_{\text{obs}}) \sim \mathcal{N}(\mu(\mathbf{X}_{\text{obs}}), \Sigma(\mathbf{X}_{\text{obs}}))$ be a multivariate Gaussian distribution conditioned on observations $\mathbf{X}_{\text{obs}}$. For any $g \in \mathbf{SO}(3)$, if $g \cdot \mu(\mathbf{X}_{\text{obs}}) = \mu(g \cdot \mathbf{X}_{\text{obs}})$ and $\Sigma(\mathbf{X}_{\text{obs}}) = \sigma^2 \mathbf{I}$, the distribution $p(\mathbf{x} \mid \mathbf{X}_{\text{obs}})$ is $\mathbf{SO}(3)$ -equivariant.*

Proof. Let d be the dimension of $\mathbf{x}$. The Probability Density Function (PDF) of $p(\mathbf{x} \mid \mathbf{X}_{\text{obs}})$ is

$$p(\mathbf{x} \mid \mathbf{X}_{\text{obs}}) = \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^\top \Sigma^{-1}(\mathbf{x} - \mu)\right), \quad (26)$$

Correspondingly, the PDF of distribution $p(g \cdot \mathbf{x} \mid g \cdot \mathbf{X}_{\text{obs}})$ is given by

$$p(g \cdot \mathbf{x} \mid g \cdot \mathbf{X}_{\text{obs}}) = \frac{1}{(2\pi)^{d/2} \det(\Sigma^g)^{1/2}} \exp\left(-\frac{1}{2}(g\mathbf{x} - \mu^g)^\top (\Sigma^g)^{-1}(g\mathbf{x} - \mu^g)\right), \quad (27)$$

where $\mu^g = \mu(g \cdot \mathbf{X}_{\text{obs}})$ and $\Sigma^g = \Sigma(g \cdot \mathbf{X}_{\text{obs}})$ denote the mean and covariance conditioned on rotated observations. By the given conditions, we have $\mu^g = g \cdot \mu$ and $\Sigma^g = \sigma^2 \mathbf{I} = \Sigma$. Thus,

$$p(g \cdot \mathbf{x} \mid g \cdot \mathbf{X}_{\text{obs}}) = \frac{1}{(2\pi)^{d/2} \det(\Sigma^g)^{1/2}} \exp\left(-\frac{1}{2}(g \cdot \mathbf{x} - \mu^g)^\top (\Sigma^g)^{-1}(g \cdot \mathbf{x} - \mu^g)\right) \quad (28)$$

$$= \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(g \cdot \mathbf{x} - g \cdot \mu)^\top (\Sigma)^{-1}(g \cdot \mathbf{x} - g \cdot \mu)\right) \quad (29)$$

$$= \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^\top g^\top \Sigma^{-1} g (\mathbf{x} - \mu)\right) \quad (30)$$

$$= \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^\top \Sigma^{-1}(\mathbf{x} - \mu)\right) = p(\mathbf{x} \mid \mathbf{X}_{\text{obs}}), \quad (31)$$

where we used that $g^\top \Sigma^{-1} g = \Sigma^{-1}$ since $\Sigma = \sigma^2 \mathbf{I}$. $\square$

Theorem B.2. *Let η denote the optimized kernel hyperparameters learned from the observations $\{\mathbf{T}^{\text{obs}}, \mathbf{X}^{\text{obs}}\}$, with the coregionalization matrix satisfying $\mathbf{B}_q = \mathbf{I}_3$. Assume observation noise is isotropic, i.e., $\Sigma = \sigma^2 \mathbf{I}_3$. For any $g \in \mathbf{SO}(3)$, $t \in \mathbf{T}^{\text{pred}}$, and rotated observations $\{\mathbf{T}^{\text{obs}}, g \cdot \mathbf{X}^{\text{obs}}\}$, the rotational equivariance of the conditional prior distribution holds as*

$$p(\mathbf{x}_i^t \mid \mathbf{T}^{\text{obs}}, \mathbf{X}_i^{\text{obs}}; \eta) = p(g \cdot \mathbf{x}_i^t \mid \mathbf{T}^{\text{obs}}, g \cdot \mathbf{X}_i^{\text{obs}}; \eta) \quad (32)$$

Proof. By the definition (25), the covariance kernel under $\mathbf{B}_q = \mathbf{I}_3$ simplifies to

$$\mathbf{K}(t, t'; \eta) = \sum_{q=1}^Q k_q(t, t'; \eta) \cdot \mathbf{I}_3, \quad (33)$$

where $k_q(t, t'; \eta)$ are base kernels parameterized by η . For vector-valued Gaussian processes, the posterior at new input is a Gaussian distribution with predictive mean and covariance given by

$$\mu_*(t_*) = \mathbf{K}(t_*, \mathbf{T}_{\text{obs}}; \eta) (\mathbf{K}(\mathbf{T}_{\text{obs}}, \mathbf{T}_{\text{obs}}; \eta) + \sigma^2 \mathbf{I}_{3N})^{-1} \bar{\mathbf{X}}_{\text{obs}}, \quad (34)$$

$$\Sigma_*(t_*) = \mathbf{K}(t_*, t_*; \eta) - \mathbf{K}(t_*, \mathbf{T}_{\text{obs}}; \eta) (\mathbf{K}(\mathbf{T}_{\text{obs}}, \mathbf{T}_{\text{obs}}; \eta) + \sigma^2 \mathbf{I}_{3N})^{-1} \mathbf{K}(t_*, \mathbf{T}_{\text{obs}}; \eta)^\top, \quad (35)$$

where $\bar{\mathbf{X}}_{\text{obs}} = [\mathbf{x}_1^\top, \mathbf{x}_2^\top, \mathbf{x}_3^\top]^\top \in \mathbb{R}^{3N}$ is the vectorized form of $\mathbf{X}_{\text{obs}}$, and N here represents the number of observed timesteps. To simplify the notation, we denote the scalar cross-covariance vector as

$$\mathbf{k}_* = [\sum_{q=1}^Q k_q(t_*, t_1; \eta), \sum_{q=1}^Q k_q(t_*, t_2; \eta), \dots, \sum_{q=1}^Q k_q(t_*, t_N; \eta)] \in \mathbb{R}^{1 \times N}, \quad (36)$$

and the scalar covariance matrix as

$$\mathbf{k}_N = \begin{pmatrix} \sum_{q=1}^Q k_q(t_1, t_1; \eta) & \sum_{q=1}^Q k_q(t_1, t_2; \eta) & \dots & \sum_{q=1}^Q k_q(t_1, t_N; \eta) \\ \sum_{q=1}^Q k_q(t_2, t_1; \eta) & \sum_{q=1}^Q k_q(t_2, t_2; \eta) & \dots & \sum_{q=1}^Q k_q(t_2, t_N; \eta) \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{q=1}^Q k_q(t_N, t_1; \eta) & \sum_{q=1}^Q k_q(t_N, t_2; \eta) & \dots & \sum_{q=1}^Q k_q(t_N, t_N; \eta) \end{pmatrix} \in \mathbb{R}^{N \times N}. \quad (37)$$

Then, the cross-covariance kernel $\mathbf{K}(t_*, \mathbf{T}_{\text{obs}}; \eta)$ and observation covariance kernel $\mathbf{K}(\mathbf{T}_{\text{obs}}, \mathbf{T}_{\text{obs}}; \eta)$ can be expressed as

$$\mathbf{K}(t_*, \mathbf{T}_{\text{obs}}; \eta) = \mathbf{I}_3 \otimes \mathbf{k}_* \in \mathbb{R}^{3 \times 3N}, \quad \mathbf{K}(\mathbf{T}_{\text{obs}}, \mathbf{T}_{\text{obs}}; \eta) = \mathbf{I}_3 \otimes \mathbf{k}_N \quad (38)$$

Let $\mathbf{K}_o = (\mathbf{k}_N + \sigma^2 \mathbf{I}_N)^{-1}$, the predictive mean simplifies to

$$\mu_*(t_*) = (\mathbf{I}_3 \otimes \mathbf{k}_*)(\mathbf{I}_3 \otimes \mathbf{K}_o) \bar{\mathbf{X}}_{\text{obs}} = \mathbf{I}_3 \otimes (\mathbf{k}_* \mathbf{K}_o) \bar{\mathbf{X}}_{\text{obs}}. \quad (39)$$

Similarly, the predictive covariance can be simplified to

$$\Sigma_*(t_*) = \sum_{q=1}^Q k_q(t_*, t_*; \eta) \cdot \mathbf{I}_3 - (\mathbf{I}_3 \otimes \mathbf{k}_*)(\mathbf{I}_3 \otimes \mathbf{K}_o)(\mathbf{I}_3 \otimes \mathbf{k}_*^\top) = \left(\sum_{q=1}^Q k_q(t_*, t_*; \eta) - \mathbf{k}_* \mathbf{K}_o \mathbf{k}_*^\top \right) \mathbf{I}_3. \quad (40)$$

Let $\sigma_* = \sum_{q=1}^Q k_q(t_*, t_*; \eta) - \mathbf{k}_* \mathbf{K}_o \mathbf{k}_*^\top$, the covariance further simplifies to

$$\Sigma_*(t_*) = \sigma_* \mathbf{I}_3. \quad (41)$$

Next, consider the rotated observations $\{\mathbf{T}_{\text{obs}}, \mathbf{X}_{\text{obs}}^g\}$. Let $\mathbf{Q} \in \text{SO}(3)$ be the rotation matrix corresponding to g . The vectorized form of $g \cdot \mathbf{X}_{\text{obs}}$ can be expressed as

$$\bar{\mathbf{X}}_{\text{obs}}^g = (\mathbf{Q} \otimes \mathbf{I}_N) \bar{\mathbf{X}}_{\text{obs}}, \quad (42)$$

Then, the predictive mean and covariance for the rotated observations are as following

$$\mu_*^g = (\mathbf{I}_3 \otimes \mathbf{k}_* \mathbf{K}_o) \bar{\mathbf{X}}_{\text{obs}}^g = (\mathbf{I}_3 \otimes \mathbf{k}_* \mathbf{K}_o) (\mathbf{Q} \otimes \mathbf{I}_N) \bar{\mathbf{X}}_{\text{obs}} \quad (43)$$

$$= (\mathbf{Q} \otimes \mathbf{k}_* \mathbf{K}_o) \bar{\mathbf{X}}_{\text{obs}} = (\mathbf{Q} \otimes \mathbf{I}_1) (\mathbf{I}_3 \otimes \mathbf{k}_* \mathbf{K}_o) \bar{\mathbf{X}}_{\text{obs}} \quad (44)$$

$$= \mathbf{Q} \mu_*, \quad (45)$$

$$\Sigma_*^g = \sigma_* \mathbf{I}_3 = \Sigma_*, \quad (46)$$

which implies that the predictive mean rotates simultaneously with the rotation of observations, while the predictive covariance remains invariant. Thus, by Lemma B.1, the posterior distribution is $\text{SO}(3)$ -equivariant, i.e.,

$$p(\mathbf{x} | \mathbf{T}_{\text{obs}}, \mathbf{X}_{\text{obs}}; \eta) = p(g \cdot \mathbf{x} | \mathbf{T}_{\text{obs}}, g \cdot \mathbf{X}_{\text{obs}}; \eta) \quad (47)$$

□

Theorem B.3. Let $p_\tau(\mathbf{X}_\tau|\mathbf{X}^{\text{obs}})$ denote the marginal probability path. If $p_0(\mathbf{X}_0|\mathbf{X}^{\text{obs}})$ is SE(3)-equivariant and $u_\tau^\theta(\mathbf{X}_\tau)$ is SO(3)-equivariant and translationally invariant, the probability path $p_\tau(\mathbf{X}_\tau|\mathbf{X}^{\text{obs}})$ is SE(3)-equivariant for all $\tau \in [0, 1]$.

Proof. Let $q_\tau(\mathbf{X}_\tau) \triangleq p_\tau(L_g(\mathbf{X}_\tau) | L_g(\mathbf{X}^{\text{obs}}))$ denote the marginal probability density under the SE(3) group actions, where the left group action is defined as $L_g(\mathbf{X}_\tau) = \mathbf{R}\mathbf{X}_\tau + \mathbf{t}$ for $g = (\mathbf{R}, \mathbf{t}) \in \text{SE}(3)$.

Since the group action L_g is time-independent, the partial derivative commutes with the group action:

$$\partial_\tau q_\tau(\mathbf{X}_\tau) = \partial_\tau p_\tau(L_g(\mathbf{X}_\tau) | L_g(\mathbf{X}^{\text{obs}})). \quad (48)$$

Next, given SO(3)-equivariance and translational invariance of u_τ^θ , i.e., $u_\tau^\theta(L_g(\mathbf{X}_\tau)) = \mathbf{R}u_\tau^\theta(\mathbf{X}_\tau)$, we have:

$$\begin{aligned} \nabla_{\mathbf{X}_\tau} \cdot (q_\tau(\mathbf{X}_\tau) u_\tau^\theta(\mathbf{X}_\tau)) &= \nabla_{L_g(\mathbf{X}_\tau)} \cdot (\mathbf{R} \cdot p_\tau(L_g(\mathbf{X}_\tau) | L_g(\mathbf{X}^{\text{obs}})) u_\tau^\theta(\mathbf{X}_\tau)) \\ &= \nabla_{L_g(\mathbf{X}_\tau)} \cdot (p_\tau(L_g(\mathbf{X}_\tau) | L_g(\mathbf{X}^{\text{obs}})) u_\tau^\theta(L_g(\mathbf{X}_\tau))). \end{aligned} \quad (49)$$

The original probability path p_τ satisfies the continuity equation, which yields

$$\partial_\tau p_\tau(L_g(\mathbf{X}_\tau) | L_g(\mathbf{X}^{\text{obs}})) + \nabla_{L_g(\mathbf{X}_\tau)} \cdot (p_\tau(L_g(\mathbf{X}_\tau) | L_g(\mathbf{X}^{\text{obs}})) u_\tau^\theta(L_g(\mathbf{X}_\tau))) = 0. \quad (50)$$

Combining Eqs. (48), (49), and (50), we find that q_τ satisfies the identical continuity equation as p_τ ,

$$\partial_\tau q_\tau(\mathbf{X}_\tau) + \nabla_{\mathbf{X}_\tau} \cdot (q_\tau(\mathbf{X}_\tau) u_\tau^\theta(\mathbf{X}_\tau)) = 0. \quad (51)$$

Then, by the SE(3)-equivariance of the prior distribution p_0 , we have

$$q_0(\mathbf{X}_0) = p_0(L_g(\mathbf{X}_0) | L_g(\mathbf{X}^{\text{obs}})) = p_0(\mathbf{X}_0 | \mathbf{X}^{\text{obs}}). \quad (52)$$

Under the smooth velocity field u_τ^θ , the Picard theorem guarantees the uniqueness of solutions to the continuity equation. Since q_τ and p_τ satisfy the same continuity equation and initial condition, we conclude:

$$q_\tau(\mathbf{X}_\tau) = p_\tau(L_g(\mathbf{X}_\tau) | L_g(\mathbf{X}^{\text{obs}})) = p_\tau(\mathbf{X}_\tau | \mathbf{X}^{\text{obs}}), \forall \tau \in [0, 1]. \quad (53)$$

This demonstrates that $p_\tau(\mathbf{X}_\tau | \mathbf{X}^{\text{obs}})$ is SE(3)-equivariant for all $\tau \in [0, 1]$. $\square$

C. More Details on Experiments

C.1. Hyperparameters

We report the detailed hyperparameters of GP-EquiFlow for all conditional trajectory prediction experiments in Table 5. we adopt the AdamW optimizer with a weight decay of $1 \cdot 10^{-12}$ and a ReduceLROnPlateau learning rate scheduler that monitors the validation loss, with a decay factor of 0.5 and a patience of 4. Validation and test evaluations are conducted every 100 training epochs for all experiments. Flow matching time steps are sampled from the distribution $\tau \sim \sqrt{\mathcal{U}(0, 1)}$ with a total of 10 sampling steps during inference. For experiments on the MD17, the batch size is set to 64 to avoid GPU out-of-memory (OOM) errors when processing molecules with a large number of atoms, e.g. aspirin and toluene. Additionally, the number of CPU processes for training the vector-valued Gaussian process priors is set to 4.

Table 5. Hyperparameters of GP-EquiFlow for all conditional trajectory prediction experiments.

	Layers	Hidden Dim	Time Emb. Dim	τ	Batch Size	LR	Epochs
N-body	6	128	32	$\sqrt{\mathcal{U}(0, 1)}, 10\text{steps}$	128	$2 \cdot 10^{-4}$	600
MD17	6	128	32	$\sqrt{\mathcal{U}(0, 1)}, 10\text{steps}$	64	$1 \cdot 10^{-4}$	1800

C.2. Choice of Kernel and Scaling Hyperparameter

The scaling hyperparameter w calibrates the predictive uncertainty of the VVGP prior. Intuitively, a smaller value of w reduces the initial discrepancy between prior-sampled trajectories and ground-truth trajectories, and it might be intuitively expected that this would lead to improved final forecasting performance. However, it is observed that excessively small values of w , e.g., 0.05, 0.1, result in severely degraded prediction accuracy, as the model tends to overfit to these overly narrow prior distributions during the training process. Conversely, a large w introduces excessive prior noise that obscures the temporal dynamic features inherent in particle trajectories, thereby increasing the learning difficulty. It is found that the model achieves consistently strong performance when w is set within the range of 0.8 to 1.5, with negligible performance variation across this interval. For consistency across all experiments reported in this work, w is fixed at 1.

For kernel function selection, a composite kernel approach is employed to adapt to different dynamical systems. For N-body systems with smooth trajectories, a combination of RBF and linear kernels is used to capture both local temporal correlations and global linear trends. For molecular dynamics systems with significant vibrational behavior, this combination is extended with a periodic kernel. This composite kernel design requires minimal manual tuning across datasets, as the VVGP training process automatically learns the relative weights of each kernel component. This allows the same kernel framework to effectively construct valid priors for different N-body systems.

C.3. Compute Resources

All training processes were conducted using 24 CPU cores and a single NVIDIA RTX 4090 GPU with 24 GB memory. Table 6 reports the total training and inference times for all experiments. During inference, we evaluated on 1000 test samples with a batch size of 100.

Table 6. Total training and inference times, rounded values.

	Training (h)	Inference (min)
Gravity	5:00	02:00
Springs	3:00	00:55
Charged	3:00	00:55
Aspirin	40:00	02:50
Benzene	30:00	01:40
Ethanol	20:00	01:30
Malonaldehyde	20:00	01:20
Naphthalene	35:00	01:15
Salicylic	30:00	02:20
Toluene	30:00	02:10
Uracil	30:00	01:45

C.4. Standard Deviations

We provide the standard deviations in Tables 7 and 8.

Table 7. Trajectory prediction on MD17 dataset, averaged over 5 runs, with standard deviations.

	Aspirin		Benzene		Ethanol		Malonaldehyde	
	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE
GP-EquiFlow	0.093±0.012	0.179±0.021	0.020±0.003	0.034±0.002	0.105±0.010	0.182±0.032	0.117±0.016	0.161±0.020
	Naphthalene		Salicylic		Toluene		Uracil	
	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE
GP-EquiFlow	0.063±0.002	0.079±0.005	0.096±0.008	0.126±0.013	0.075±0.012	0.109±0.019	0.065±0.006	0.085±0.008

Table 8. Trajectory prediction on N-Body systems, averaged over 5 runs, with standard deviations

	Gravity		Spring		Charged	
	ADE	FDE	ADE	FDE	ADE	FDE
GP-EquiFlow	0.105 ± 0.002	0.274 ± 0.02	0.0019 ± 0.00005	0.0039 ± 0.00007	0.068 ± 0.001	0.152 ± 0.002

D. Additional Results

D.1. Additional Results on Different Sampling Steps

We further provide Negative Log-Likelihood (NLL) on the *Spring*, *Gravity*, *Benzene*, and *Toluene* datasets with different sampling steps during inference. The results are presented in Table 9. GP-EquiFlow achieves accurate prediction even with only 2 steps and reaches the best overall performance on most datasets with 10 steps.

Table 9. Negative Log-Likelihood (NLL) with different Euler sampling steps during inference. Results are averaged over 5 independent runs. The best results are **bolded**.

Dataset	Euler-2	Euler-5	Euler-10	Euler-20	Euler-50
Spring	933.70	-274.06	-282.83	-285.71	-287.14
Gravity	-48.76	-49.33	-56.98	-44.48	-30.23
Benzene	-124.61	-132.38	-141.57	-132.21	-130.38
Toluene	-69.87	-76.63	-74.26	-72.27	-70.47

D.2. Ablation Studies with Fixed Kernel Hyperparameters

Table 10. Average Displacement Error (ADE) and Final Displacement Error (FDE) for trajectory prediction on N-body and MD17 datasets with different kernel types, averaged over 5 runs. The best results are **bolded** and the second best are underlined.

Dataset	Spring		Gravity		Charged		Aspirin		Ethanol		Malonaldehyde		Salicylic		Uracil	
	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE
GeoTDM	<u>0.0030</u>	<u>0.0079</u>	0.256	0.613	0.110	0.258	0.107	0.193	0.115	0.209	0.107	0.176	0.083	0.120	0.074	0.099
RBF	0.0098	0.0141	0.102	0.256	0.058	0.125	0.092	0.156	0.095	0.170	<u>0.097</u>	<u>0.153</u>	<u>0.077</u>	<u>0.101</u>	0.070	0.089
Linear	0.0099	0.0166	0.353	0.745	0.153	0.347	0.234	0.364	0.286	0.392	<u>0.248</u>	<u>0.369</u>	0.166	0.218	0.149	0.167
Matern	0.0054	0.0105	<u>0.104</u>	<u>0.257</u>	0.097	0.200	0.097	<u>0.167</u>	0.134	0.219	0.088	0.141	0.070	0.091	<u>0.066</u>	0.083
Periodic	0.0132	0.0246	0.425	0.796	0.212	0.411	0.296	0.443	0.314	0.393	0.289	0.462	0.189	0.256	0.097	0.121
GP-EquiFlow	0.0019	0.0039	0.105	0.274	<u>0.068</u>	<u>0.152</u>	<u>0.093</u>	0.179	<u>0.105</u>	<u>0.182</u>	0.117	0.161	0.096	0.126	0.065	<u>0.085</u>

We conduct additional ablation studies to evaluate the performance of fixed and untrained kernel functions for prior construction. In these experiments, all kernel hyperparameters are manually predefined and remain constant throughout training and inference processes.

The four evaluated kernel functions are:

$$\begin{aligned}
k_{\text{RBF}}(t_1, t_2) &= \sigma_{\text{rbf}}^2 \exp\left(-\frac{(t_1 - t_2)^2}{2l_{\text{rbf}}^2}\right) \\
k_{\text{Matern}}(t_1, t_2) &= \sigma_{\text{matern}}^2 \left(1 + \frac{\sqrt{3}|t_1 - t_2|}{l_{\text{matern}}}\right) \exp\left(-\frac{\sqrt{3}|t_1 - t_2|}{l_{\text{matern}}}\right) \\
k_{\text{Linear}}(t_1, t_2) &= \sigma_{\text{linear}}^2 t_1 t_2 \\
k_{\text{Periodic}}(t_1, t_2) &= \sigma_{\text{periodic}}^2 \exp\left(-\frac{2 \sin^2\left(\pi \frac{|t_1 - t_2|}{P}\right)}{l_{\text{periodic}}^2}\right)
\end{aligned}$$

where l denotes the length scale, σ denotes the amplitude, and P denotes the period of the periodic kernel.

In our experiments, the specific hyperparameters used for each kernel are $l_{\text{rbf}} = 0.5$, $\sigma_{\text{rbf}} = 1.0$; $l_{\text{matern}} = 2.0$, $\sigma_{\text{matern}} = 1.0$; $\sigma_{\text{linear}} = 1.0$; $l_{\text{periodic}} = 1.0$, $P = 10.0$, $\sigma_{\text{periodic}} = 1.0$.

The results are presented in Table 10. It is found that prediction performance is sensitive to both the choice of kernel type and the predefined hyperparameters. In this work, we do not conduct an extensive hyperparameter search, and further performance improvements can be achieved through more fine-grained tuning of kernel parameters and exploration of kernel combinations.

Overall, the RBF and Matern kernels with predefined hyperparameters achieve the best performance across most datasets. This superior performance can be attributed to the consistency of the prior distribution throughout the training process, which simplifies the learning task of the downstream flow matching model. Notably, fixed kernel approaches require strict consistency between the kernel configuration used during training and inference. In contrast, learned prior methods such as GeoTDM and GP-EquiFlow do not suffer from this limitation.